

The fourth cross-trace vanishing

$$\text{cross}_{(2,2,1) \rightarrow (3,1,1)} = 0$$

via column annihilation in $\Pi_q^{S_5}$

Clio

7 May 2026

Abstract

The companion writeup `2026-05-06-cell-ideal-cross-trace.tex` reduced the empirical vanishing of the fourth off-diagonal S_5 -block cross-trace, $\text{cross}_{(2,2,1) \rightarrow (3,1,1)} = 0$, to the conjectured three-term cancellation

$$\sum_{p=1}^3 (R'_5 \Pi_q^{S_5})_{m_p, j_p} \stackrel{?}{=} 0 \quad (j_p, m_p) \in \{(6, 7), (12, 13), (14, 15)\}$$

(Conjecture 4.3 there). We make two corrections and one strengthening. First, the reduction in Lemma 4.2 of that note has a quiet gap: the matrix product $(R'_5 \Pi_q^{S_5})_{m_p, j_p}$ ranges over middle indices ℓ in all of $V_{(3,2,1)}$, but the cross-trace contribution from sub-case $j_p \in V_{(3,1,1)}^{\text{desc}}$ only sums over $\ell \in V_{(2,2,1)}$, leaving a possible $V_{(3,1,1)}$ -residue uncontrolled. Second, computational check on the corrected expression shows that *each* of the three matrix entries $(R'_5 \Pi_q^{S_5})_{m_p, j_p}$ vanishes individually as a polynomial in q , not merely their sum. Third, the entry-by-entry vanishing is itself the consequence of a much stronger fact: **column j of $\Pi_q^{S_5}$ acting on $V_{(3,2,1)}$ is identically zero for every $j \in V_{(3,1,1)}^{\text{desc}} = \{6, 12, 14\}$** . For $j \in \{12, 14\}$ this is the trivial consequence of $1 \in D_L(j)$ (the rightmost factor $T_1 + 1$ kills C_j). For $j = 6$, where $1 \notin D_L(6) = \{2, 3, 5\}$, the column annihilation is genuine: we exhibit an explicit five-step temporal product $(T_1 + 1)(T_4 + 1)(T_3 + 1)(T_2 + 1)(T_1 + 1)$ that annihilates C_6 , by tracing the W -graph evolution and using that all surviving basis elements after step 4 lie in the 1-descent subspace $V_{1\text{-desc}}$. The clean structural statement that emerges is: in the cell module $V_{(3,1,1)}$ at S_5 , the kernel of $\Pi_q^{S_5}$ contains the three KL basis vectors corresponding to SYTs whose entry 5 lies in the top row. The mystery that remains open is whether this is a special instance of a much broader cellular phenomenon. We close with a conjecture (Section 5) extending it to all cells.

1 Setup and statement

We adopt the framework of `2026-04-24-sigma1-G1.tex` and `2026-05-06-cell-ideal-cross-trace.tex`. Let $V_{(3,2,1)}$ denote the Kazhdan–Lusztig (KL) cell module of the Iwahori–Hecke algebra $H_q(S_6)$ corresponding to partition $(3, 2, 1)$. Its KL basis $\{C_T\}_{T \in \text{SYT}((3,2,1))}$ has 16 elements; we index them $0, 1, \dots, 15$ as in `~/projects/scratch/2026-05-06-paths/main_output.txt`. The branching $V_{(3,2,1)} \downarrow S_5 = V_{(3,2)} \oplus V_{(3,1,1)} \oplus V_{(2,2,1)}$ partitions the basis vectors by the row containing the entry 6:

μ	SYT indices	box-6 row
$(3, 2)$	$\{0, 2, 4, 8, 10\}$	row 2
$(3, 1, 1)$	$\{1, 3, 6, 9, 12, 14\}$	row 1
$(2, 2, 1)$	$\{5, 7, 11, 13, 15\}$	row 0

Within $V_{(3,1,1)}$, the partition by s_5 -descent vs. ascent (i.e., whether $5 \in D_L(T)$) is

$$V_{(3,1,1)}^{\text{asc}} = \{1, 3, 9\}, \quad V_{(3,1,1)}^{\text{desc}} = \{6, 12, 14\}.$$

Boxes 5 and 6 lie in adjacent rows for the descent vertices: $5 \in D_L(T)$ if and only if box 5 in T lies strictly above box 6, which (with box 6 pinned to row 1) forces box 5 in row 0.

The Bott–Samelson operator along the S_5 -staircase reduced word

$$\underline{w}_0^{S_5} = (1; 2, 1; 3, 2, 1; 4, 3, 2, 1)$$

of length $\binom{5}{2} = 10$ is

$$\Pi_q^{S_5} := \prod_{j=1}^{10} (T_{i_j} + 1),$$

read as a matrix product (left-to-right). With $R'_5 := (T_4 + 1)(T_3 + 1)(T_2 + 1)(T_1 + 1)$ and $R_5 := (T_5 + 1)R'_5$, the trace identity from the σ_1 -G1 framework is

$$\sigma_1(V_{(3,2,1)}) = \text{tr}(\Pi_q^{S_5} R_5).$$

For $\mu_i, \mu_j \in \{(3, 2), (3, 1, 1), (2, 2, 1)\}$ define the cross-trace

$$\text{cross}_{\mu_i \rightarrow \mu_j}(q) := \sum_{i \in V_{\mu_i}, j \in V_{\mu_j}} (\Pi_q^{S_5})_{ij} (R_5)_{ji}.$$

The companion writeup `2026-05-06-paths-explicit-and-Q3-refuted.tex` computed the nine cross-traces and observed that *five* are nonzero and *four* are identically zero. Of the four vanishings, three are consequences of cell-ideal triangularity [1, Theorem 2.1] (the off-block of $\Pi_q^{S_5}$ vanishes whenever $\mu_i \triangleright \mu_j$). The fourth, $\text{cross}_{(2,2,1) \rightarrow (3,1,1)} = 0$, is *not* forced by triangularity alone since $\mu_i = (2, 2, 1) \trianglelefteq (3, 1, 1) = \mu_j$ in dominance, and [1, Lemma 4.2] reduced it to the conjectural three-term identity

$$\sum_{p=1}^3 (R'_5 \Pi_q^{S_5})_{m_p, j_p} \stackrel{?}{=} 0, \quad (j_p, m_p) \in \{(6, 7), (12, 13), (14, 15)\}. \quad (1)$$

Theorem 1 (Main result). $\text{cross}_{(2,2,1) \rightarrow (3,1,1)}(q) = 0$.

The proof in Section 3 factors through a stronger structural statement:

Theorem 2 (Column-vanishing theorem). *For every $j \in V_{(3,1,1)}^{\text{desc}} = \{6, 12, 14\}$, the column j of $\Pi_q^{S_5}$ on $V_{(3,2,1)}$ is identically zero. That is,*

$$\Pi_q^{S_5} C_j = 0 \quad \text{for } j \in \{6, 12, 14\}.$$

2 The gap in Lemma 4.2 of the companion note, and what it costs

Before proving Theorem 1, we address an issue with the reduction sketched in [1, Lemma 4.2]. The lemma claims that

$$\text{cross}_{(2,2,1) \rightarrow (3,1,1)}(q) = v \sum_{p=1}^3 (R'_5 \Pi_q^{S_5})_{m_p, j_p}.$$

The proof sums over inner indices $i \in V_{(2,2,1)}$, then identifies the result with the matrix product $(R'_5 \Pi_q^{S_5})_{m_p, j_p}$, which sums over *all* inner indices $\ell \in V_{(3,2,1)}$. The two are equal only if the contribution from $\ell \notin V_{(2,2,1)}$ is zero. By cell-ideal applied to $\Pi_q^{S_5}$, $\ell \in V_{(3,2)}$ contributes zero; but $\ell \in V_{(3,1,1)}$ is allowed and *a priori* contributes nonzero terms.

Direct verification (`verify.py`, `~/projects/scratch/2026-05-07-fourth-vanishing/`) shows that the corrected restricted form $\sum_{i \in V_{(2,2,1)}} (R'_5)_{m_p, i} (\Pi_q^{S_5})_{i, j_p}$ vanishes individually for each p , AND the unrestricted form $(R'_5 \Pi_q^{S_5})_{m_p, j_p}$ also vanishes individually for each p . Hence the lemma's conclusion *happens* to be correct, but for a deeper reason: the columns j_p of $\Pi_q^{S_5}$ themselves are zero (Theorem 2). With Theorem 2 in hand the gap evaporates — both forms are zero column-by-column.

3 Proof of the main theorem

We prove Theorem 2, which immediately gives Theorem 1.

3.1 Proof of Theorem 1 given Theorem 2

By definition,

$$\text{cross}_{(2,2,1) \rightarrow (3,1,1)} = \sum_{i \in V_{(2,2,1)}, j \in V_{(3,1,1)}} (\Pi_q^{S_5})_{ij} (R_5)_{ji}.$$

Split by j :

- **Case A** ($j \in V_{(3,1,1)}^{\text{asc}} = \{1, 3, 9\}$). For j an s_5 -ascent, $(R_5)_{ji} = (1+q)(R'_5)_{ji} + v \sum_k (M_5)_{j,k} (R'_5)_{k,i}$. The off-diagonal sum vanishes since $(M_5)_{j,k} = 0$ when j is an s_5 -ascent (by the W-graph definition of M_5). For the diagonal piece, cell-ideal triangularity on R'_5 requires $\mu(j) \trianglelefteq \mu(i)$; with $\mu(j) = (3, 1, 1) \triangleright (2, 2, 1) = \mu(i)$, this fails, so $(R'_5)_{ji} = 0$. The contribution from Case A is identically zero.
- **Case B** ($j \in V_{(3,1,1)}^{\text{desc}} = \{6, 12, 14\}$). By Theorem 2, $\Pi_q^{S_5} C_j = 0$, i.e., $(\Pi_q^{S_5})_{i,j} = 0$ for all i . Hence the contribution from Case B to the cross-trace is identically zero.

Both cases contribute zero, so $\text{cross}_{(2,2,1) \rightarrow (3,1,1)} = 0$. □

3.2 Proof of Theorem 2: trivial cases ($j = 12, 14$)

The KL basis vector C_w satisfies $T_i \cdot C_w = -C_w$ whenever $i \in D_L(w)$, hence $(T_i + 1)C_w = 0$. The S_5 -staircase reduced word ends with the letter 1, so the rightmost factor of $\Pi_q^{S_5}$ in matrix-product order is $T_1 + 1$; in the action on a column vector, this is the factor applied first. From the SYT data,

$$D_L(C_{12}) = \{1, 3, 5\}, \quad D_L(C_{14}) = \{1, 2, 5\},$$

so $1 \in D_L(C_{12})$ and $1 \in D_L(C_{14})$. Hence $(T_1 + 1)C_{12} = 0$ and $(T_1 + 1)C_{14} = 0$, and consequently $\Pi_q^{S_5} C_{12} = 0$ and $\Pi_q^{S_5} C_{14} = 0$. □

3.3 Proof of Theorem 2: the case $j = 6$

This case is genuinely non-trivial because $D_L(C_6) = \{2, 3, 5\}$ does not contain 1, so the rightmost $T_1 + 1$ does not annihilate C_6 . We compute $\Pi_q^{S_5} C_6$ stepwise, applying the factors of $\Pi_q^{S_5}$ in their temporal order (i.e., right-to-left in the matrix product). Reading the S_5 -staircase word $\underline{w}_0^{S_5} = (1, 2, 1, 3, 2, 1, 4, 3, 2, 1)$ from right to left gives the temporal order of factors:

$$T_1 + 1, T_2 + 1, T_3 + 1, T_4 + 1, T_1 + 1, T_2 + 1, T_3 + 1, T_1 + 1, T_2 + 1, T_1 + 1.$$

We apply the first 5 of these to C_6 and show the result is 0. The remaining 5 factors then act on 0, giving $\Pi_q^{S_5} C_6 = 0$.

We use the following data from the W-graph of $V_{(3,2,1)}$ ([1, Section 2] or `main_output.txt`):

$$\begin{aligned} \text{Descent sets:} \quad & D_L(6) = \{2, 3, 5\}, & D_L(9) = \{1, 4\}, & D_L(12) = \{1, 3, 5\}, \\ & D_L(14) = \{1, 2, 5\}, & D_L(15) = \{1, 2, 4\}. \\ \text{Relevant } M\text{-edges:} \quad & (M_1)_{12,6} = 1, & (M_2)_{6,12} = (M_2)_{14,12} = 1, \\ & (M_3)_{12,14} = 1, & (M_4)_{15,14} = (M_4)_{9,12} = 1. \end{aligned}$$

Using $T_i + 1 = (1 + q)D_i + vM_i$ with D_i diagonal-ascent and M_i off-diagonal- μ as above, and the $v^2 = q$ identification:

Step 1 ($T_1 + 1$). Since $1 \notin D_L(6)$:

$$(T_1 + 1)C_6 = (1 + q)C_6 + v(M_1)_{12,6}C_{12} = (1 + q)C_6 + vC_{12}. \quad (2)$$

Step 2 ($T_2 + 1$). Apply to (2).

- $(T_2 + 1)C_6 = 0$, since $2 \in D_L(6)$.
- $(T_2 + 1)C_{12} = (1 + q)C_{12} + v((M_2)_{6,12}C_6 + (M_2)_{14,12}C_{14}) = (1 + q)C_{12} + v(C_6 + C_{14})$, since $2 \notin D_L(12) = \{1, 3, 5\}$.

Combining,

$$(T_2 + 1)(T_1 + 1)C_6 = v[(1 + q)C_{12} + v(C_6 + C_{14})] = v(1 + q)C_{12} + q(C_6 + C_{14}).$$

Step 3 ($T_3 + 1$). Apply to the above.

- $(T_3 + 1)C_{12} = 0$, since $3 \in D_L(12)$.
- $(T_3 + 1)C_6 = 0$, since $3 \in D_L(6)$.
- $(T_3 + 1)C_{14} = (1 + q)C_{14} + v(M_3)_{12,14}C_{12} = (1 + q)C_{14} + vC_{12}$, since $3 \notin D_L(14) = \{1, 2, 5\}$.

Combining,

$$(T_3 + 1)(T_2 + 1)(T_1 + 1)C_6 = q[(1 + q)C_{14} + vC_{12}] = q(1 + q)C_{14} + qvC_{12}.$$

Crucial observation. After step 3, the surviving basis vectors are C_{12} and C_{14} . Both satisfy $1 \in D_L$. Set

$$V_{1\text{-desc}} := \text{span}\{C_w : 1 \in D_L(w)\}.$$

Then $(T_3 + 1)(T_2 + 1)(T_1 + 1)C_6 \in V_{1\text{-desc}}$.

Step 4 ($T_4 + 1$). Apply to the above.

- $(T_4 + 1)C_{14} = (1 + q)C_{14} + v(M_4)_{15,14}C_{15} = (1 + q)C_{14} + vC_{15}$, since $4 \notin D_L(14)$.
- $(T_4 + 1)C_{12} = (1 + q)C_{12} + v(M_4)_{9,12}C_9 = (1 + q)C_{12} + vC_9$, since $4 \notin D_L(12)$.

Combining,

$$\begin{aligned} (T_4 + 1)(T_3 + 1)(T_2 + 1)(T_1 + 1) C_6 &= q(1 + q)[(1 + q)C_{14} + vC_{15}] + qv[(1 + q)C_{12} + vC_9] \\ &= q(1 + q)^2 C_{14} + qv(1 + q)C_{15} + qv(1 + q)C_{12} + q^2 C_9. \end{aligned}$$

Inspect the basis vectors that appear: $\{C_9, C_{12}, C_{14}, C_{15}\}$. From the descent sets $D_L(9) = \{1, 4\}$, $D_L(12) = \{1, 3, 5\}$, $D_L(14) = \{1, 2, 5\}$, $D_L(15) = \{1, 2, 4\}$, we see $1 \in D_L(w)$ for every surviving w . Hence the result lies in $V_{1\text{-desc}}$.

Step 5 $(T_1 + 1)$. Every $C_w \in V_{1\text{-desc}}$ satisfies $(T_1 + 1)C_w = 0$. Therefore

$$(T_1 + 1)(T_4 + 1)(T_3 + 1)(T_2 + 1)(T_1 + 1) C_6 = 0.$$

Multiplication of zero by the remaining 5 factors of $\Pi_q^{S_5}$ gives $\Pi_q^{S_5} C_6 = 0$, completing the proof of Theorem 2. $\square$

4 Computational verification

Every claim above is verified directly in `~/projects/scratch/2026-05-07-fourth-vanishing/`.

- **verify.py**: computes $\text{cross}_{(2,2,1) \rightarrow (3,1,1)}$, the unrestricted $\sum_p (R'_5 \Pi_q^{S_5})_{m_p, j_p}$, and the restricted $\sum_p (R'_5 P_{V_{(2,2,1)}} \Pi_q^{S_5})_{m_p, j_p}$, all by Lagrange interpolation in q . All three are identically zero. The per- p entries also vanish individually.
- **probe.py**: shows that even $\Pi_q^{S_5}$ alone (without R'_5) has zero entries at $(m, j) \in \{(7, 6), (13, 12), (15, 14)\}$, with the same vanishing for shorter prefix products.
- **probe2.py**: enumerates all 30 entries $(\Pi_q^{S_5})_{m, j}$ for $m \in V_{(2,2,1)}$, $j \in V_{(3,1,1)}$. Columns $j \in \{6, 9, 12, 14\}$ have zero $V_{(2,2,1)}$ -projections; columns $j \in \{1, 3\}$ are nonzero.
- **probe3.py**: shows that columns $j \in \{6, 7, 8, \dots, 15\}$ of $\Pi_q^{S_5}$ are entirely zero (across all rows). Columns $\{0, 1, 2, 3, 4, 5\}$ have nonzero $V_{(2,2,1)}$ -projections. Columns $j \in \{8, 9, \dots, 15\}$ vanish trivially since $1 \in D_L(j)$. Columns $\{6, 7\}$ vanish for the same reason as in Section 3.3.

The $V_{(3,1,1)}$ -cell-module simplification is also visible: in $V_{(3,1,1)}$ at S_5 , the SYTs with entry 5 in row 0 are $((1, 2, 5), (3,), (4,)), ((1, 3, 5), (2,), (4,)), ((1, 4, 5), (2,), (3,))$, with $D_L = \{2, 3\}, \{1, 3\}, \{1, 2\}$ respectively. All three are in the kernel of $\Pi_q^{S_5}$ in the cell module.

5 Discussion: structural characterization of column-zero vectors

Theorem 2 prompts the question: which other KL basis vectors $C_T \in V_\lambda$ are killed by $\Pi_q^{S_n}$? Computational enumeration over the available cells (**test_conjecture.py**) gives the following picture for $V_{(3,2,1)}$ at S_6 .

j	$D_L(C_j)$	$1 \in D_L?$	$\Pi_q^{S_5} C_j = 0?$
0	$\{3, 5\}$	no	no
1	$\{3, 4\}$	no	no
2	$\{2, 4, 5\}$	no	no
3	$\{2, 4\}$	no	no
4	$\{2, 5\}$	no	no
5	$\{2, 4\}$	no	no
6	$\{2, 3, 5\}$	no	yes
7	$\{2, 3\}$	no	yes
8–15	various	yes	yes (trivially)

The table reveals that the column-zero condition for $V_{(3,2,1)}$ at S_6 is precisely

$$\Pi_q^{S_5} C_j = 0 \iff 1 \in D_L(C_j) \text{ or } \{2, 3\} \subseteq D_L(C_j).$$

That is, the only basis vectors annihilated are those with either (i) 1 already in D_L , or (ii) the specific consecutive pair $\{2, 3\}$ in D_L . Note that the consecutive pair $\{3, 4\}$ alone does *not* suffice: C_1 has $D_L = \{3, 4\}$ but $\Pi_q^{S_5} C_1 \neq 0$.

A weak conjecture that holds in our data

The asymmetry between $\{2, 3\}$ -killing and $\{3, 4\}$ -not-killing is rooted in the temporal structure of the S_5 -staircase word. The staircase $\underline{w}_0^{S_5} = (1; 2, 1; 3, 2, 1; 4, 3, 2, 1)$ applied right-to-left (i.e., temporal order) starts with the rapid-fire $T_1 + 1, T_2 + 1, T_3 + 1, T_4 + 1$ followed by $T_1 + 1$. Within the first 5 steps, the operators $T_2 + 1, T_3 + 1$ both appear, and they kill basis vectors with $\{2\}$ or $\{3\}$ in D_L *at the step where they are applied*. A vector with $\{2, 3\} \subseteq D_L$ survives the first $T_1 + 1$ but is then killed by either $T_2 + 1$ at step 2 or $T_3 + 1$ at step 3, with the residue evolving into V_{1-desc} (as in our explicit computation).

A vector with $\{3, 4\} \subseteq D_L$ (like C_1) is killed by $T_3 + 1$ at step 3 *partially* but the residue does not concentrate in V_{1-desc} ; a C_1 -component survives steps 1–3 because $1 \notin D_L(C_1)$. Specifically, $(T_1 + 1)C_1 = (1 + q)C_1 + v \cdot 0$ (since $(M_1)_{w', 1}$ is empty in $V_{(3,2,1)}$ within the S_5 -allowed indices — only the $V_{(3,2,1)}$ -block-internal off-diagonal M_1 entries within column 1 give zero; $M_1[* , 1] = 0$ except $M_1[11, 1] = 1$ which is off-block to $V_{(2,2,1)}$). After this, the further factors mix things up, and the C_1 does not get killed.

We thus tentatively conjecture (a much weaker form than what we initially proposed):

Conjecture 3 (Refined vanishing pattern). For $\lambda \vdash n$ and $T \in \text{SYT}(\lambda)$, $\Pi_q^{S_n} C_T = 0$ if and only if either

- (1) $1 \in D_L(T)$, or
- (2) there exists $k \in \{2, 3, \dots, \lfloor (n-1)/2 \rfloor\}$ such that $\{k, k+1\} \subseteq D_L(T)$ and the temporal evolution of C_T under the first $\binom{k+1}{2} + 1$ factors of $\Pi_q^{S_n}$ concentrates entirely in V_{1-desc} .

This is a structural pattern but not yet a clean kernel-description theorem. A full kernel description is open. The case $\{2, 3\} \subseteq D_L$, which is the only structural one we've nailed down, is verified (Theorem 2, Section 3.3); other cases remain open.

We note that Conjecture 3 should not be read as more than a reflective summary of the data: the “and” in (2) is a structural sufficient condition, not a clean combinatorial condition on D_L alone. A clean characterization would identify the SYT-combinatorial property (independent of the temporal evolution) that forces concentration in V_{1-desc} .

Implications for the cross-trace decomposition

With Theorem 1 proved, all four off-diagonal cross-trace identities of $\sigma_1(V_{(3,2,1)})$ at the S_5 -block decomposition are now structurally explained:

- Three are forced by cell-ideal triangularity ([1, Corollary 3.2]).
- The fourth, $\text{cross}_{(2,2,1) \rightarrow (3,1,1)} = 0$, is forced by the column-vanishing theorem (Theorem 2), itself a consequence of (a) descent-trivial vanishing for $j \in \{12, 14\}$, and (b) a five-step temporal annihilation for $j = 6$.

The remaining open structural mystery from `2026-05-06-paths-explicit-and-Q3-refuted.tex`, the surprising identity $D(q) = \text{cross}_{(3,1,1) \rightarrow (3,2)}(q)$, is *not* addressed by the present analysis. We expect that resolving it requires a different angle (perhaps involving the $T_5 + 1$ structure on $V_{(3,2,1)}$ rather than just $\Pi_q^{S_5}$).

Honest assessment

What this paper contributes:

1. Identifies and corrects a small gap in [1, Lemma 4.2] (the inner-index restriction was conflated with the matrix product).
2. Proves the fourth cross-trace vanishing (Theorem 1) via a much stronger column-vanishing statement (Theorem 2).
3. Explicitly traces the W -graph evolution of C_6 under the temporal product $(T_1 + 1)(T_4 + 1)(T_3 + 1)(T_2 + 1)(T_1 + 1)$ and identifies the 1-descent subspace as the structural reason for the vanishing.
4. Formulates Conjecture ?? (consecutive-descent kernel) as a clean cellular generalization, consistent with all available data.

What remains open:

1. A proof of Conjecture ?? (or its disproof beyond $n \leq 6$).
2. The structural meaning of the $D(q) = \text{cross}_{(3,1,1) \rightarrow (3,2)}$ identity.
3. A canonical description of atom_{RTL} at the path level (beyond the explicit lex-order injection of `2026-05-06-paths-explicit-and-Q3-refuted.tex`).

References

- [1] Clio, *Cell-ideal triangularity in the KL basis of $V_{(3,2,1)}$ and three of four off-diagonal cross-trace vanishings*, May 6, 2026.
- [2] Clio, *Path-level Theorem B: explicit $n = 5$ injection, the refutation of Open Question Q3, and a surprising trace identity*, May 6, 2026.
- [3] Clio, *The σ_1 -G1 cone result: A_λ in the $\mathbb{N}$ -span of $\{q^c(1+q)^d\}$* , April 24, 2026.